



# How trees influence slope stability during cyclones: field and experimental evidence

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## Abstract

The dual role of trees in affecting slope stability during cyclones is complex and remains underexplored. This study investigates how tree roots can both stabilize slopes through mechanical reinforcement and destabilize them by facilitating rapid infiltration. We examine 95 cyclone-induced landslides and conduct a series of field surveys, laboratory tests, and dye tracer experiments on slopes covered by *Pinus massoniana*. Results show that tree roots significantly increase soil porosity and permeability, creating preferential flow paths that enhance subsurface moisture during storms. At the same time, root reinforcement provides additional cohesion to the shallow soil layer. However, the combined effects of wind loading and concentrated infiltration in the rooted area contribute to failure surfaces at the maximum rooting depth and the bedrock-soil interface. These findings highlight the importance of incorporating both the stabilizing and destabilizing effects of trees into landslide risk assessments under cyclonic conditions.

**Keywords** Cyclone-induced landslides · Trees-covered slope · Root reinforcement · Trees' detrimental effects · Potential sliding surfaces

## 1 Introduction

In the context of global climate change, extreme cyclone events frequently occur (Elsner et al. 2008; Patricola and Wehner 2018). Risk assessment of cyclone-induced landslides is now a primary concern in coastal hill regions worldwide (Chang et al. 2014; Saito et al. 2014;

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Cogan et al. 2018). In the coastal regions, many of the cyclone-induced landslides occur at approximately the same time when cyclones strike and are clustered in trees-covered slopes (Zhuang et al. 2022), indicating a clustered spatial distribution and potentially distinctive failure mechanisms. Therefore, it is essential to investigate how tree-related mechanical and hydrological processes contribute to the initiation of landslides during cyclones.

It is recognized that trees affect slope stability via a series of mechanical and hydrological processes. For the mechanical process, root reinforcement plays a major role in stabilizing the slope (Schwarz et al. 2013; Wu 2013; Masi et al. 2021; Lann et al. 2024; Song et al. 2024). The presence of roots leads to an increase in soil shear strength through the tensile, compressive, and shear resistance of roots, along with frictional interaction with the soil (Cislaghi et al. 2019). This mechanism makes trees widely used to control landslides worldwide (Cohen and Schwarz 2017). However, there is controversy over the mechanical process of trees on slope stability (Simon and Collison 2002; Zhuang et al. 2022). During a cyclone event, the wind load acting on trees can transmit a load of over 10 kN to the slope (Zhuang et al. 2022, 2023). When the wind blows downhill, the detrimental effect of wind load combined with the surcharge of trees (Cislaghi et al. 2018) acts as a driving force to reduce slope stability (Emadi-Tafti and Ataie-Ashtiani 2019). In some cases, the contribution of wind load can outweigh the beneficial impacts of root reinforcement.

In terms of the hydrological processes, the impacts of trees again separate as beneficial and detrimental parts. The tree crown creates an “umbrella effect” in rainfall, intercepting part of precipitation and attenuating the rainwater penetration in the slope (Gonzalez-Ollauri and Mickovski 2017; Sidle and Ziegler 2017). Root-water uptake and transpiration process absorb moisture, desiccates the soil surrounding the roots, and thus generates matric suction (Zhu and Zhang 2015; Ng et al. 2016). Both beneficial hydrological processes reduce the water content within the slope, but are suggested to show a weak impact in intense rainfall (McGuire et al. 2016; Gonzalez-Ollauri and Mickovski 2017). Furthermore, plant roots often provide preferential flow paths within the slope (Zhuang et al. 2023). Compared with the uniform matrix flow, preferential flow describes rapid water evolution within the soil matrix via the macro pores network (Allaire et al. 2009). Decayed tree roots work as cracks, providing long channels to accelerate the water penetration within the slope (Guo et al. 2019). For living roots, there are gaps of millimeters at the root-soil interface (Ghestem et al. 2011; Li et al. 2023). The connected roots can make a network of flowing paths that favor rapid water movement. Due to the biological activities, root systems also create large pores and facilitate the formation of wormholes in the reinforced area (Bodner et al. 2014; Nimmo 2021). These hydrological processes are influential to the mechanical and permeability properties of soil, and thus affect the slope stability during a cyclone event.

The impact of trees on slope stability is multi-faced, complex, and heavily related to the meteorological condition and vegetation species. During a cyclone event, roots play a vital role in linking trees and the slope, such as transmitting wind load and changing soil properties (Zhuang et al. 2023; Lann et al. 2024). The contribution of roots is therefore worth investigating because it might be an important factor for the slope failure and development of sliding surfaces. To our knowledge, the comprehensive impacts of trees on cyclone-induced landslides are rarely reported.

With this purpose, we investigated 95 cyclone-induced landslides and summarized their general features. A landslide case covered by typical plant species was selected to experimentally investigate the distribution of roots within the slope. Subsequently, further labo-

ratory tests and dye tracer tests were conducted to elucidate the impacts of roots on soil properties. The primary objective of this study is to investigate how trees affect slope stability during cyclones, focusing on both mechanical (e.g., root reinforcement and wind load transmission) and hydrological (e.g., preferential infiltration and changes in soil properties) processes. We further identify the location of potential sliding surfaces, particularly with respect to the relative depths of roots and soil. The findings provide insights into future risk assessment of cyclone-induced landslides in forest-covered regions.

## 2 Study area

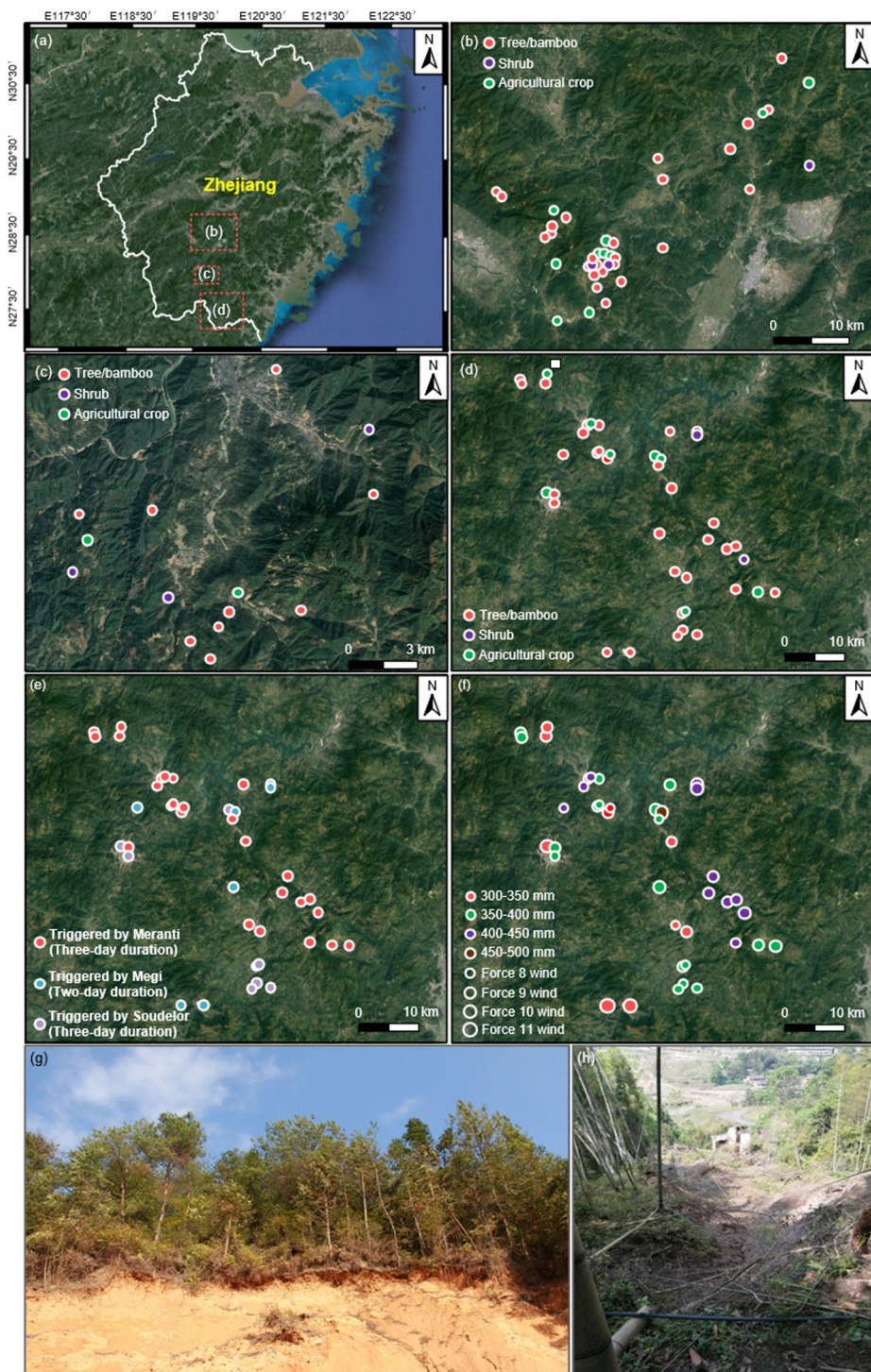
Southeast coastal China, which faces the Pacific Ocean, is one of the regions most severely affected by cyclones (Xiao et al. 2011; Huang et al. 2022). In the area, Zhejiang province (Fig. 1a) is frequently struck by cyclones and the generated landslides caused mass casualties and economic loss.

Zhejiang Province is dominated by mountainous and hilly terrain, covering approximately 70.4% of its total area, with an overall southwest-to-northeast descending gradient forming a step-like topography. The southern part is characterized by steep mountains and deeply incised valleys, with average elevations exceeding 800 m, whereas the central region consists mainly of low mountains and hills below 500 m. The northern and eastern regions are largely coastal plains with low relief, having average elevations of 2–5 m. The cyclone-affected areas are primarily located in the eastern and southern mid- to low-mountain belts, with elevations ranging from 200 m to 1200 m. Volcanic rocks are widely developed and serve as the principal material source for natural hazards in the area. Most of these volcanic formations date to the Mesozoic era, dominated by continental rhyolite–dacite assemblages. Acidic volcanic rocks such as rhyolitic lava, welded tuff, and tuff account for 61–78% of the total volcanic rock. The province lies in a subtropical monsoon climate zone with abundant rainfall. The annual precipitation ranges from 980 mm to 2000 mm, generally increasing from north to south and with higher amounts in mountainous areas. The southeastern mountainous region receives the highest precipitation, forming a rainfall center of approximately 2000 mm per year, and is the area most severely affected by typhoons (Zhang 2016).

In this study, we investigated 95 cyclone-induced landslides in the area and summarized their features. The locations of the identified landslide cases are presented in Fig. 1a–d.

### (1) General characteristics

Among the 95 cyclone-induced landslides, 90 cases are soil landslides, and the number of rock slides and soil-rock mixed slides is 4 and 1, respectively. A significant portion, 91%, of these events have a depth of <3 m and only 2 cases exceeding 10 m. According to the landslide classification proposed by National Research Institute for Earth Science and Disaster Prevention (depth < 10 m is shallow landslide, depth > 10 is deep-seated landslide), cyclone-induced landslides in the study area are primarily shallow soil landslides. Furthermore, the slopes are mostly characterized by a two-layer structure, signifying that a layer of weathering-formed loose soil deposits overlays the bedrock. Two prevalent sliding surfaces are observed in situ. 57% of the identified landslides occurred at the bedrock–soil interface, while 29% occurred at weak layers in the overburden soil layer. Notably, the field survey



**Fig. 1** Investigated Cyclone-induced landslides **a–f** Locations and characteristics of landslides; **g–h** Landslide cases occurred in trees-covered slopes

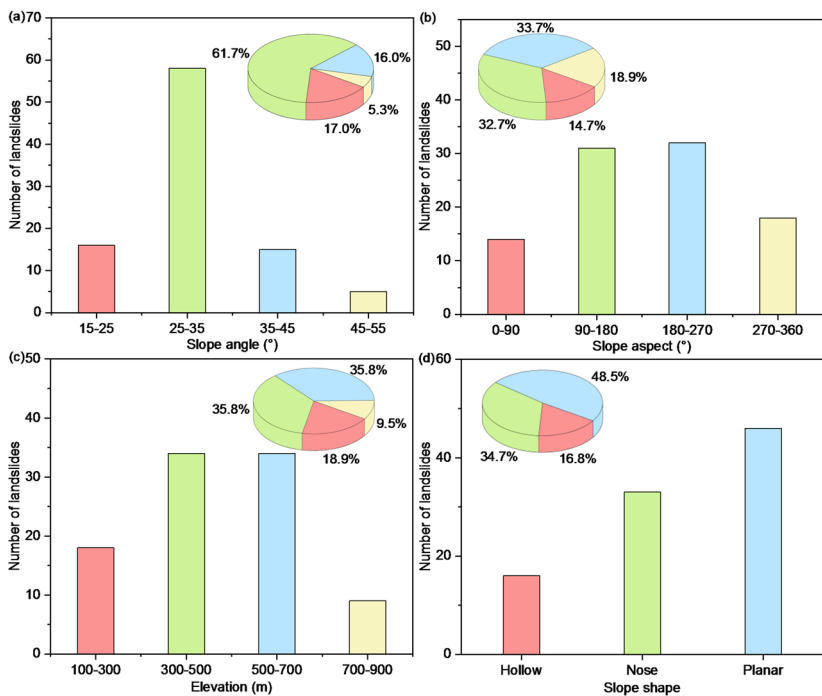


was conducted immediately after the cyclone as an emergency investigation. Due to the large number of landslides and limited time, only the depth of the sliding surface was measured, while the area and volume of each case were not recorded in detail.

## (2) Geological and geomorphological characteristics

Cyclone-induced landslides in the study area are most prevalent in slopes of angles ranging from 25°–35°, accounting for 61.7% of the total amount (Fig. 2a). The average slope angle of investigated cases is 30.4°, with a few landslides occurring on slopes with angles below 20° and above 40°. The landslide frequency shows a correction with the slope aspect (Fig. 2b). 66.4% of cyclone-induced landslides occurred on slopes with aspects ranging from 90° to 270°, indicating a higher landslide risk in south-facing slopes. Only one-third of landslides occurred on north-facing slopes.

In terms of elevation, the identified cases are concentrated in slopes of an elevation ranging from 300 to 700 m a.s.l. (Fig. 2c). Slopes in this area are characterized by low to middle mountains. The steep terrain and frequent human activities provide conditions to facilitate the landslide initiation. Moreover, we further characterize the slope using the curvature value, and categorize them as convex (nose), planar and concave (hollow) slopes (Ohlmacher 2007). Results indicate that cyclone-induced landslides are clustered in planar and convex slopes (Fig. 2d). Planar-shaped slopes account for 48.5% of the total, followed by convex upward slopes at 34.7%. Only 16.8% of landslides occurred in concave upward slopes.



**Fig. 2** Geological and geomorphological characteristics of identified landslides. **a** Slope angle; **b** Slope aspect; **c** Elevation; **d** Slope shape

### (3) Vegetation characteristics

Cyclone-induced landslides occurred on slopes exhibiting high vegetation coverage. Various plant species (including *Pinus massoniana*, *Cunninghamia lanceolata*, *Phyllostachys edulis*) are identified on the slopes (Fig. 1b–d), with trees and bamboo (*Phyllostachys edulis*) being the most prevalent (Fig. 1e and f), followed by agricultural crops (Fig. 1g) and shrubs (Fig. 1h). 73.7% of the landslides occurred on slopes covered by trees and bamboo of large frontal areas, indicating a potential relationship between landslide initiation and trees. Trees widely distributed in the study area are pines (*Pinus massoniana*) and firs (*Cunninghamia lanceolata*).

### (4) Meteorological conditions

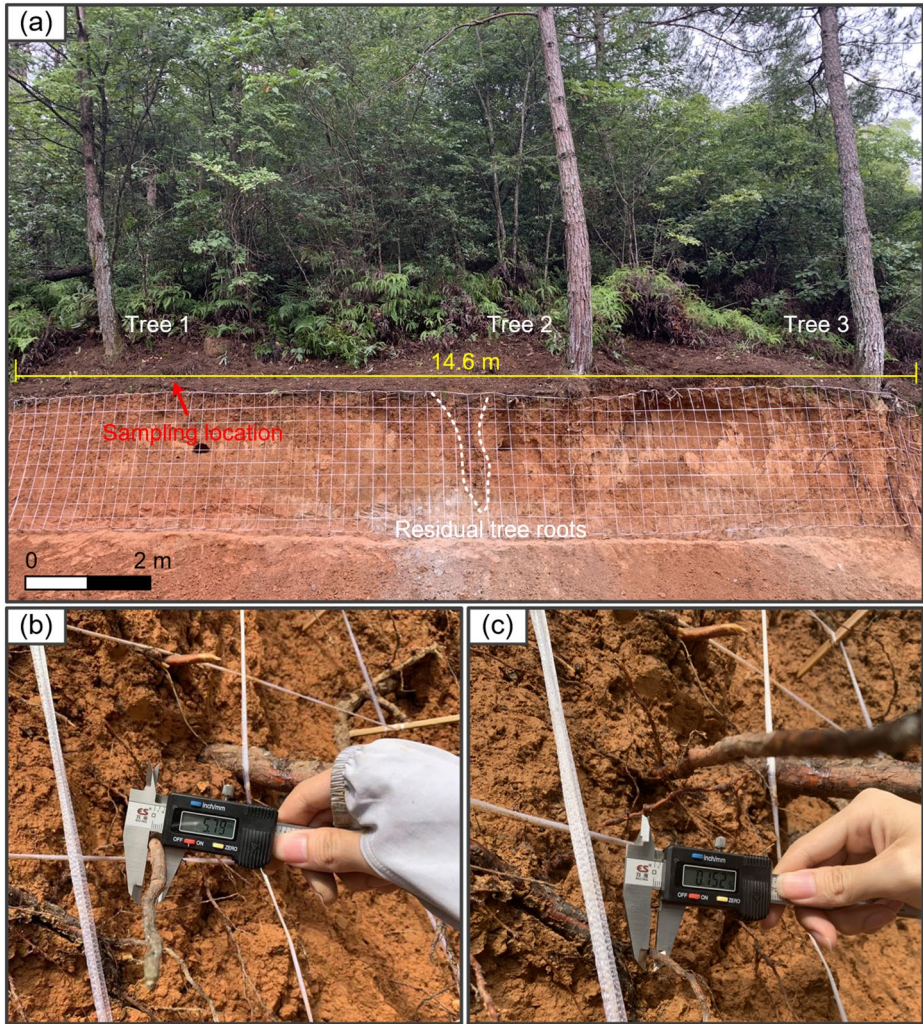
In this study, 41 identified landslide cases in Wenzhou (Fig. 1e and f) have meteorological data records of precipitation and maximum wind force (Beaufort wind force). The cumulative precipitation at locations of all the landslides exceeded 300 mm with a concentration range of 300–450 mm, indicating the importance of intense rainfall to landslide initiation. Meteorological data of wind force illustrates the great contribution of wind load. All the identified landslides are located within the radii of seven-scale and ten-scale wind circles. 17% of the landslides experienced a peak wind force of over 10 (Fig. 1d). In terms of temporal distribution, cyclone-induced landslides occurred at approximately the same time as cyclone strikes. 88% of the landslides occurred on the day of cyclone landing or the following day, corresponding to the peak period of wind force and rainfall. The time-lag effect is weak.

## 3 Methods and data

### 3.1 Roots distribution measurements

Here we selected a cyclone-induced shallow landslide case with a depth of 2.5–3.0 m and a two-layer structure (upper soil layer and lower bedrock) (Fig. 3a). The maximum wind force at the landslide site reached Beaufort scale 10. The slope angle is 28° and the landslide occurred at the bedrock-soil interface. Trees covered on the slope are *Pinus massoniana* (Masson's pine), which are universally distributed in the study area (Zhuang et al. 2023). Trees have an average height of 20 m and a diameter at breast height (DBH) of 25–30 cm (Wang 2014; Zhuang et al. 2022).

We excavated a vertical profile at the landslide crown (Fig. 3a). The profile measures 14.6 m in width and 2 m in depth, both of which are determined by the extent of tree root spread and the characteristics of the slope. Totally 3 standing trees are included in the profile (named tree 1–3 from left to right) and there are residual taproots of a dead tree. To investigate the distribution of roots, the profile is divided into 730 meshes (size: 20 cm×20 cm) with plastic strips. The root number and the diameter of every single root are measured for each mesh (Fig. 3b and c). The distribution of roots is described by two indices: root



**Fig. 3** Root distribution measurement. **a** Profile excavation and mesh division; **b–c** Root diameter measurement

number density (*RND*) and root area ratio (*RAD*) (Ali et al. 2023; Deljouei et al. 2023; Li et al. 2023).

*RND* is defined as the ratio of the root number (*n*) to the soil cross-sectional area ( $A_s$ ):

$$RND = \frac{n}{A_s} \quad (1)$$

*RAD* is defined as the ratio of the cross-sectional area of roots ( $A_r$ ) to the soil ( $A_s$ ):

$$RAD = \frac{A_r}{A_s} = \frac{\sum_{i=1}^n \frac{\pi d_i^2}{4}}{A_s} \quad (2)$$

where  $d_i$  is the diameter of measured roots. In this study,  $RND$  and  $RAD$  are separately measured in each mesh, thus the soil cross-sectional area  $A_s = 20 \times 20 = 400 \text{ cm}^2$ .

### 3.2 Laboratory tests on soil properties

To investigate the impacts of roots on soil properties, samples of root-reinforced soil were collected at the landslide crown using ring knives. We collected samples at 0.2 m intervals from the ground surface to 2.0 m in depth, including soil with and without roots. Notably, due to the limitation of experimental instruments, the sampling location is  $\sim 1 \text{ m}$  from the tree to avoid the influence of coarse roots (Fig. 3a).

The properties of root-reinforced soil are measured following the United States Department of Agriculture (USDA 2004). The dry unit weight is measured using the ring knife method and the oven-drying method. The specific gravity ( $G_s$ ) of the soil samples is determined using the pycnometer test. The root-soil sample is a multiphase mixture of soil particles, roots, water and void, thus the measurement of porosity ( $n$ ) refers to Jotisankasa and Sirirattanachat (2017). The porosity of root-soil composites is described as:

$$n = \frac{V_v}{V} = \frac{V - V_s - V_r}{V} = \frac{V - \frac{W_s}{G_s \gamma_w} - \frac{W_r}{G_r \gamma_w}}{V} \quad (3)$$

where  $V (\text{cm}^3)$  is the total volume, ( $V_v$ ,  $V_s$ ,  $V_r$ ) is the volume of void, soil and root, respectively ( $\text{cm}^3$ ), ( $W_s$ ,  $W_r$ ) is the weight of soil particles and roots (g), ( $G_s$ ,  $G_r$ ) is the specific gravity of soil particles and roots,  $\gamma_w (\text{g/cm}^3)$  is water density.

The porosity of rootless soil samples is described as:

$$n = \frac{V_v}{V} = \frac{V - V_s}{V} = \frac{V - \frac{W_s}{G_s \gamma_w}}{V} \quad (4)$$

When the slope fails in a cyclone event, the soil mass often has high water content. Here direct shear tests are applied on undisturbed samples to measure the mechanical parameters in the saturated state. The shear strength parameters are obtained according to the Mohr-Coulomb criterion:

$$\tau = c + \sigma \tan \varphi \quad (5)$$

where  $\tau (\text{kPa})$  is the shear strength,  $\sigma = \sum_{i=1}^n \gamma_i h_i (\text{kPa})$  represents the normal stress,  $\gamma_i (\text{kN/m}^3)$  and  $h_i (\text{m})$  are the unit weight and thickness of the  $i$ th soil layer,  $c (\text{kPa})$  is cohesion and  $\varphi (^\circ)$  is the friction angle.

Further falling-head permeability tests are applied to measure the saturated hydraulic conductivity ( $K_s$ ) of the undisturbed samples. The  $K_s$  (m/s) values of both root-soil composites and rootless soil are calculated as (Sandoval et al. 2017):



$$K_s = 2.3 \frac{aL}{At} \log_{10} \frac{H_1}{H_2} \quad (6)$$

where  $a$  (m<sup>2</sup>) is the cross-section area of the cube,  $A$  (m<sup>2</sup>) and  $l$  (m) are the cross-section area and length of the sample,  $t$  (s) is the interval time,  $h_1$  (m) and  $h_2$  (m) are the initial and final pressure height, respectively. Each experiment was conducted in triplicate, and the average values were used in this study.

### 3.3 Dye tracer test and image analysis

#### (1) Dye tracer test

Here dye tracer tests are conducted to elucidate the potential relationship between roots and preferential flow, also to clarify the type of preferential flow path in trees-covered slopes in the study area.

A *Pinus massoniana* with a DBH of ~30 cm was selected to do the analysis (Fig. 4). A rectangular wooden frame was vertically embedded into the soil at a depth of 20 cm. The length and width of the frame are 100 cm and 80 cm, respectively. According to the requirements of the local authorities, the experiment cannot cause significant damage to the trees. Therefore, in order to protect the trees and considering the direction of root development, the trees was not placed at the center of the dye application area. We selected methylene blue for the experiment because of its visibility and low toxicity (Nandy and Venkatesh 2010; Mills 2012; Zhang and Xu 2016), and made the dye solution concentration of 4 g/L. The dye solution was uniformly sprayed with an intensity of 50 mm/h, lasting for 3 h (equal to 150 mm rainfall). No ponding or runoff happened during the experiment. After spraying, the frame was covered by a plastic film to prevent evaporation and rainfall.

After 24 h, when the dyeing process was completed, we removed the plastic film and started to excavate the profiles. To protect the tree (keep it alive, according to Chinese laws), two profiles were excavated on both sides of the plot, as presented in Fig. 4. The excavation depth depends on the depth of dyed soil and roots. After the excavation, the soil profiles were trimmed with a knife and brush. The dyed profiles were photographed by the camera for the subsequent analysis.

#### (2) Image analysis

The Java-based image-processing program ImageJ<sup>®</sup> (Rasband 2007), combined with Photoshop was applied to analyze the dyed profiles. The contrast and brightness of photographs were firstly adjusted in Photoshop, which helped to distinguish the dyed region. Subsequently, the images were binarized and processed according to the threshold brightness value. The dyed region was represented by black, and the ImageJ<sup>®</sup> software calculated the associated area (Kramers et al. 2009). In this study, the Java-based automatic method was applied to investigate the variation of preferential infiltration in depth. The images were vertically automated layered and then batch processed, so that the dyed area of each layer was



**Fig. 4** Dye tracer test (a) The initial condition of the tree; (b) the dyed area and the location excavated profiles; (c) View of the excavated Profile 1

obtained. The automated grid partitioning technique significantly improves the efficiency and accuracy of image recognition.

The infiltration path in the soil profile is characterized by 2 indices (Bargués Tobella et al. 2014; Luo et al. 2019): dye coverage (*DC*) and preferential flow fraction (*Pr-fr*). *DC* is the proportion of the dyed area to the total area:

$$DC = \frac{D}{D + ND} \times 100\% \quad (7)$$

where *D* and *ND* are the dyed area and undyed area, respectively.

*Pr-fr* represents the proportion of preferential infiltration and is described as:

$$\text{Pr-fr} = \left(1 - \frac{\text{UniFr} \cdot W}{\text{TSA}}\right) \times 100\% \quad (8)$$

where *UniFr* is the depth of uniform infiltration, defined as the depth at which the dye coverage (*DC*) decreases to below 80%, *W* is width of the profile, and *TSA* is the total dyed area.

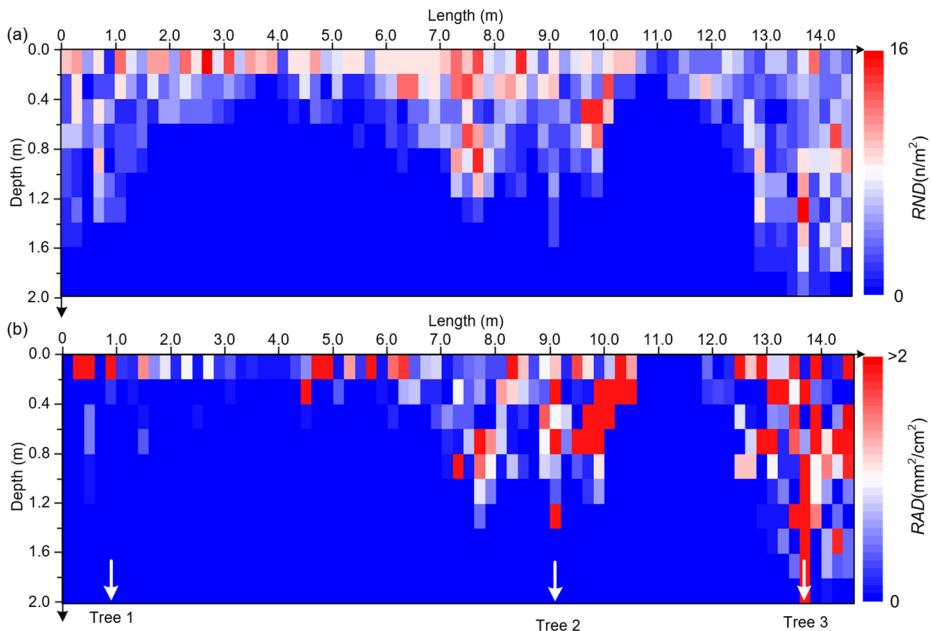
## 4 Results

### 4.1 Distribution of plant roots

Spatial distribution of plant roots is presented in Fig. 5. There are 1409 roots accounted in the profile. The maximum rooting depth reaches 2 m.

According to the measurement of root number density (*RND*) (Fig. 5a), the maximum *RND* value in a single mesh is 400/m<sup>2</sup> (16 roots in a single mesh), which is located at depths of 0–0.2 m. In terms of vertical distribution, plant roots show a uniform distribution in 0–0.2 m in depth. At depths > 0.4 m, the number of roots greatly decreases, and many meshes are rootless. In the profile, plant roots have two sources: trees (*Pinus massoniana*) and fern (*Pteris multifida*). Fern roots primarily distribute in 0–0.4 m in depth, especially at depths of 0–0.2 m. Therefore, we find a uniform root distribution close to the ground surface. In the deeper area, the fern roots disappear and the identified roots are attributable to trees.

In terms of horizontal distribution, roots are concentrated near the trees. At depths > 0.6 m, roots are clustered at the three standing trees and the one dead tree (residual roots), while the



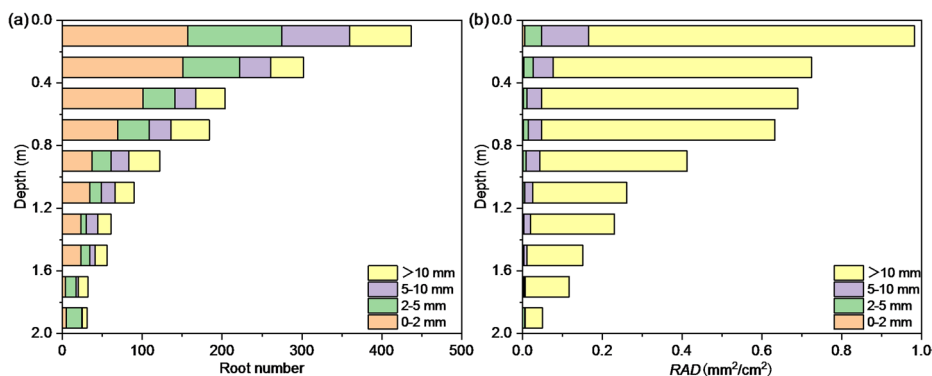
**Fig. 5** Distribution of roots in the profile. **a**RND; **b**RAD

other areas are nearly rootless. Notably, the maximum rooting depth of 2 m is identified at 13.8 m from the origin of the profile, corresponding to tree 3. Tree 3 is tightly closed to the excavated profile and the taproots are exposed, indicating the maximum rooting depth and root type. Compared with trees of a plate-like root system, *Pinus massoniana* in the study area indicates a strong taproot system but relatively sparse lateral roots. Its rooting depth is significantly larger than fern, thus the roots have a larger impact area and are influential to slope stability during a cyclone event.

Figure 5b shows the distribution of root area ratio (*RAD*). Similar to *RND*, large *RAD* values are again located at the three standing trees and the dead tree. Fern roots, which are concentrated close to the ground surface, are numerous but have a small diameter. Therefore, the total cross-section area of fern roots is limited and blue areas are identified between trees. For the *Pinus massoniana*, taproots of tree 3 are exposed in the excavated profile, so that large *RAD* values occur throughout the profile. Tree 1 is ~1 m away from the excavated profile and taproots are hidden inside the slope. According to the *RND* and *RAD* measurements, the *RAD* values of tree 1 roots are smaller than the other two trees because the exposed roots have a small diameter.

To further investigate the distribution of plant roots, we separate the roots into four diameter classes: fine roots (0–2 mm), small roots (2–5 mm), medium roots (5–10 mm) and coarse roots (> 10 mm). Results indicate that both the number and cross-sectional area of roots greatly decreases with depth (Fig. 6). The root number reaches 437 at depths of 0–0.2 m and 302 at depths of 0.2–0.4 m, a rather high value, but drops to a value of < 100 at depths > 1 m. Though plant roots in the excavated profile are primarily fine and small roots, the proportion of coarse roots begins to increase at a depth > 0.4 m. This phenomenon matches the *RND* distribution (Fig. 5a) that plant roots at a depth > 0.4 m are mostly attributable to trees.

The *RAD* of different diameter classes shows a similar variation trend with root number (Fig. 6b). The *RAD* reaches the peak value of ~1.0 mm<sup>2</sup>/cm<sup>2</sup> at depths of 0–0.2 m, and drops to a value of 0.04 mm<sup>2</sup>/cm<sup>2</sup> at depth of 1.8–2.0 m. Though plant roots in the excavated profile are primarily fine and small roots, both classes weakly contribute to the cross-section area because of the small diameter. The *RAD* proportion of coarse roots accounts for over 80% in the entire profile.



**Fig. 6** Vertical distribution of roots in different classes. **a** Root number; **b** *RAD*



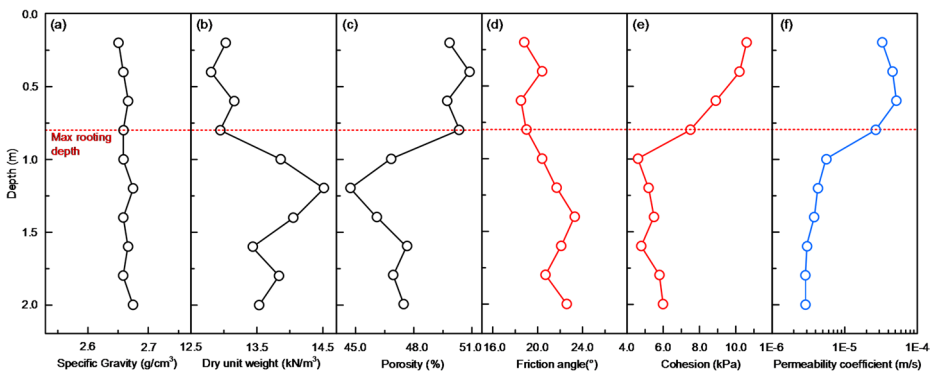
## 4.2 Properties of root-soil composites

### 4.2.1 Physical properties

Physical properties of the root-soil samples obtained from laboratory tests are presented in Fig. 7a–c. Soil in the study area is silty clay. At the sampling location, the maximum rooting depth is  $\sim 0.8$  m. Therefore, samples at depths of 0–0.8 m are root-soil composites and samples at depths  $>0.8$  m are rootless. Experimental results show a nearly constant specific gravity ( $G_s$ ) in depth (Fig. 7a), ranging from 2.65 to 2.68  $\text{g/cm}^3$ , indicating a uniform soil layer. Furthermore, the dry density (Fig. 7b) and porosity (Fig. 7c) of samples exhibit an opposite variation. A turning point is observed at 0.8 m in depth, corresponding to the maximum rooting depth. At depths of 0–0.8 m, the dry density and porosity show a value of 12.8–13.1  $\text{kN/m}^3$  and 49.7%–50.9%, respectively. At depths  $>0.8$  m, the dry density and porosity show a value of 13.5–14.5  $\text{kN/m}^3$  and 44.7%–47.6%, respectively. The variations of dry density and porosity illustrate the influence of roots on physical properties of the soil. Roots facilitate the formation of pores within the soil. This phenomenon, combined with the low root density, results in a lower density and higher porosity of root-soil composites than rootless soil.

### 4.2.2 Mechanical and hydrological properties

Mechanical parameters of samples at different depths are presented in Fig. 7d and e. The friction angle remains stable at  $18.5\text{--}23.3^\circ$ , increasing slightly with depth. However, the cohesion presents a different variation. A turning point again occurs at the maximum rooting depth (0.8 m). The cohesion of samples shows a value of 7.5–10.6 kPa at depths of 0–0.8 m, and a value of 4.6–6.0 kPa at depths  $>0.8$  m. In the rootless area, the cohesion slightly increases with depth and has a mean value of 5.3 kPa. If we make 5.3 kPa as the benchmark to represent the cohesion arising from the soil, root reinforcement at depths of 0–0.8 m provides an additional cohesion of 2.2–5.3 kPa. Such additional cohesion greatly increases slope stability. Laboratory test results suggest plant roots in the study area stabilize the slope by providing cohesion but are less influential to the friction angle.

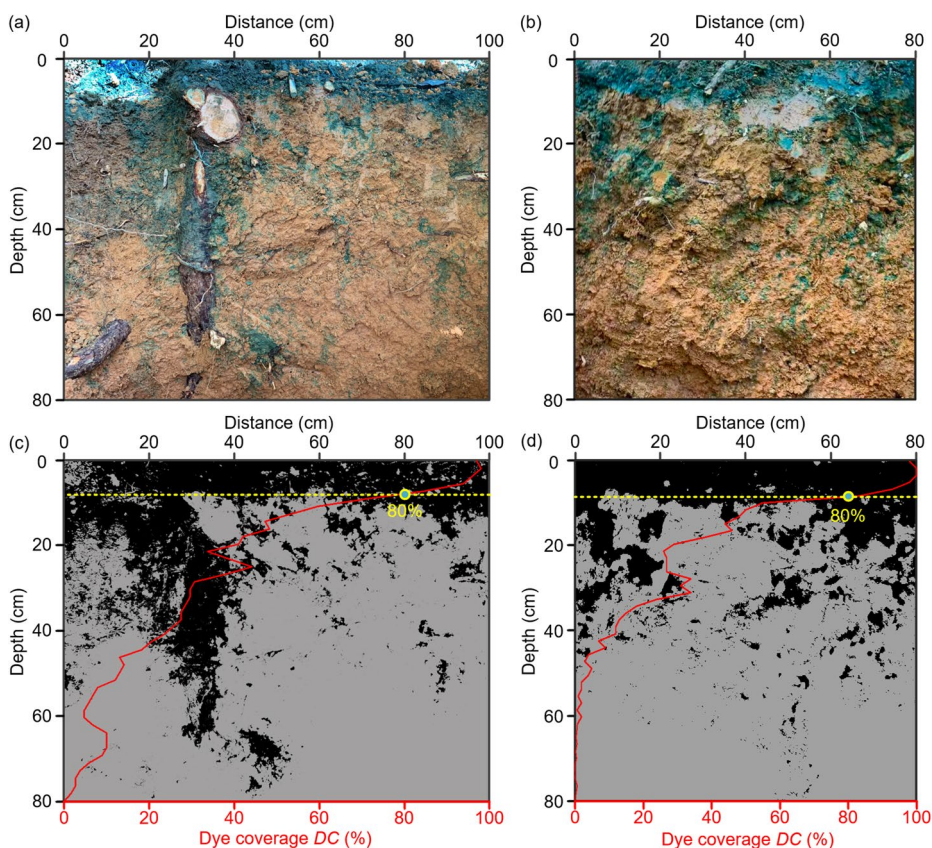


**Fig. 7** Impacts of roots on soil properties. **a** Specific Gravity; **b** Dry unit weight; **c** Porosity; **d** Friction angle; **e** Cohesion; **f** Permeability coefficient

Figure 7f shows the laboratory results of the saturated hydraulic conductivity ( $K_s$ ). The  $K_s$  value ranges from  $2.9 \times 10^{-6}$  to  $5.2 \times 10^{-5}$  m/s and greatly decreases with depth. At depths of  $>0.8$  m, the  $K_s$  ranges from  $2.9 \times 10^{-6}$  to  $5.6 \times 10^{-6}$  m/s with an average value of  $3.74 \times 10^{-6}$  m/s. However, for the root-soil composites, the  $K_s$  value reaches  $2.7 \times 10^{-5}$ – $5.2 \times 10^{-5}$  m/s, nearly 10 times the rootless soil. The presence of roots enlarges the soil porosity, increasing the soil permeability, and thus accelerates water penetration during a cyclone event.

### 4.3 Preferential infiltration paths

Laboratory tests suggest that roots greatly magnify the soil permeability, but the dominant preferential flow path in the trees-covered slope is unclear. Figure 8 shows the results of dye tracer tests and image analysis. The excavated depth, which is dependent on the dyed depth, is 80 cm in both profiles. To investigate the variation of dye coverage (DC) with depth, the profiles are vertically automated layered with an interval of 2 cm (40 layers in total). Image analysis of both profiles indicates a decreasing trend of DC with depth. The DC value reaches over 80% at depths of 0–8 cm, indicating a uniformly shallow infiltration



**Fig. 8** Image-analysis results of dye tracer tests. **a–b** Original images of dyes profiles; **c–d** Analysis of dye coverage (DC) using ImageJ®

in the area. At depths  $> 8$  cm, the *DC* rapidly declines and the movement of dye solution is dominated by preferential infiltration.

According to Eq. 8, the total dyed area *TSA* of profile 1 is  $2710 \text{ cm}^2$  and the dyed area of preferential flow reaches  $1986 \text{ cm}^2$ , indicating a *Pr-fr* value of 73.3% (Fig. 8a and c). The contribution of preferential infiltration is almost three times the uniform infiltration. Both the dye tracer results and the measured permeability coefficients underscore the significant impact of preferential flow on infiltration rates. Therefore, it is essential to account for the influence of root systems when assessing real-world infiltration behavior. Though there are few cracks and wormholes in the excavated profile, the preferential flow is clustered at tree roots, especially the coarse roots. At depths  $> 8$  cm (Fig. 8c), especially depths  $> 20$  cm, preferential flow primarily happens at 0–40 cm from the origin of the profile, corresponding to the root-reinforced area. The dye solution moves radially in all directions centered on the coarse roots. Gaps at root-soil interfaces are the primary preferential flow paths in the profile. For profile 2 (Fig. 8b and d), the *DC* drops to a very low value (nearly 0) at 65 cm in depth. The total dyed area *TSA* and the dyed area of preferential flow reach  $1603 \text{ cm}^2$  and  $976 \text{ cm}^2$ , respectively, representing a *Pr-fr* value of 60.9%. Though the *Pr-fr* value is slightly lower than that in profile 1, preferential flow is the primary source of moisture within the slope. Compared with profile 1, the number and cross-section area of roots in profile 2 is small and lacks coarse roots, thus the dyed area (preferential flow paths) is sporadically distributed in the profile.

To further elucidate the contribution of tree roots to preferential flow, the root data in the profiles is collected with a vertical interval of 10 cm. The relationship between tree roots and preferential flow is presented in Fig. 9. In both profiles, the *RAD* of roots and dye coverage (*DC*) show a similar variation, greatly dropping with depth. In profile 1, due to the coarse roots, the maximum *RAD* value reaches  $3.5 \text{ mm}^2/\text{cm}^2$ , which is much higher than in profile 2. Therefore, a larger *DC* value is observed in profile 1. In profile 2, both the *RAD* and *DC* drop to a low value at depths  $> 40$  cm. In this area, soil is almost rootless and preferential flow is weak. Our test results suggest preferential flow paths in the study area primarily arise from tree roots, more precisely, the gaps between root-soil interfaces. The coarse roots are greatly influential to the rapid water penetration because of their large cross-section areas.

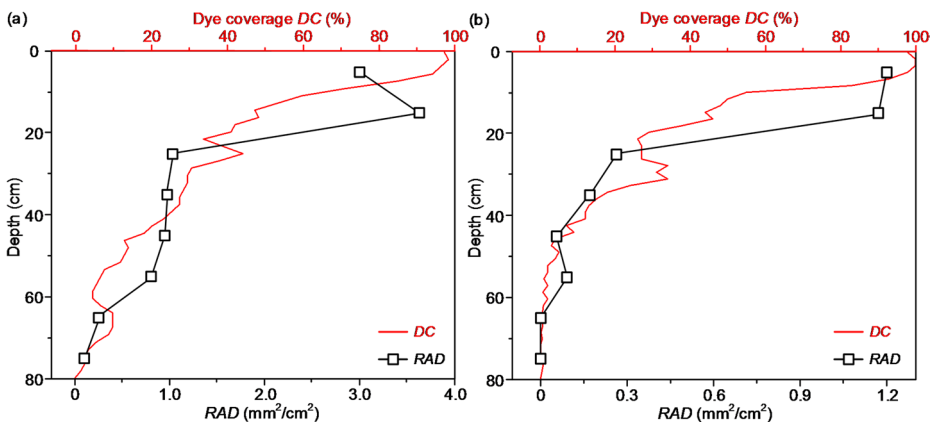


Fig. 9 Relationship between roots and dye coverage. a Profile 1; b Profile 2

#### 4.4 Slope stability analysis

To further investigate the impacts of trees on landslide initiation, we conducted a slope stability analysis using the experimental data and the limit equilibrium method proposed by Zhuang et al. (2022). The slope was modeled as an infinite slope with a constant inclination of  $28^\circ$ , comprising an upper soil layer 3 m thick underlain by bedrock. The maximum rooted depth is 2 m, and the tree distance is 5 m. The friction angle ( $21^\circ$ ), cohesion of the soil (5 kPa), and the additional strength of the roots arising from the mean measured value (4 kPa). Given the high rainfall intensity during cyclones and the enhanced infiltration induced by tree roots, the slope can be saturated with a short duration. Therefore, the saturated hydraulic conductivity was adopted for the calculations. The factor of safety is calculated with:

$$\begin{cases} F_s = \frac{c_s + (\gamma h \cos \beta + W_t) \cos \beta \tan \phi + c_r}{(\gamma h \cos \beta + W_t) \sin \beta + F_w \cos \beta / L} & h \leq h_{root} \\ F_s = \frac{c_s + (\gamma h \cos \beta + W_t) \cos \beta \tan \phi}{(\gamma h \cos \beta + W_t) \sin \beta} & h > h_{root} \end{cases} \quad (9)$$

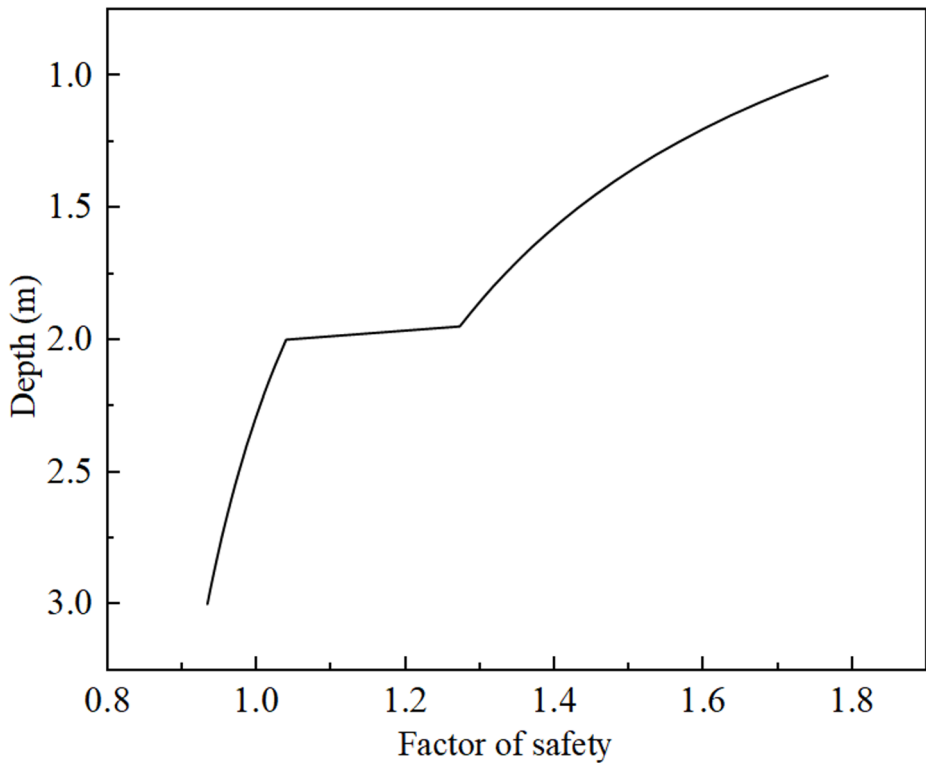
where  $c_s$  is the soil cohesion,  $c_r$  is the additional cohesion of roots,  $\gamma$  is the soil unit weight,  $h$  is the soil depth,  $h_{root}$  is the maximum rooting depth,  $\beta$  is the slope angle,  $\phi$  is the friction angle,  $W_t$  is the weight of a tree and  $F_w$  is the wind load. Here, we suggest the wind blows downhill and the calculated  $F_s$  is the lowest value. According to Zhuang et al. (2022), for the mature *Pinus massoniana* with a DBH of  $\sim 30$  cm, the load transmitted by a single tree under a Force 10 cyclone can reach 10 kN. This value is applied for the analysis.

The calculated results are presented in Fig. 10. The factor of safety decreases continuously with increasing depth. A sudden drop occurs at a depth of 2 m below the ground surface, corresponding to the location of maximum rooting depth. In this region, the root reinforcement disappears, leading to a significant reduction in soil strength and a sharp decline in the factor of safety. Below 2 m, the factor of safety continues to decrease, reaching a minimum value of 0.93 at the bedrock-soil interface. Notably, the preferential infiltration effect provided by trees is largely confined to the root zone; during cyclonic rainfall, transient saturation can readily develop near root tips, causing the reinforced soil layer to quickly become saturated and prone to failure at the maximum rooting depth. These results suggest that, for cyclone-induced landslides, the maximum rooting depth and the bedrock-soil interface - where soil properties undergo abrupt changes - represent potential internal sliding surfaces.

## 5 Discussion

Trees are widely used to control landslides because of root reinforcement and beneficial hydrological processes (Gonzalez-Ollauri and Mickovski 2017; Masi et al. 2021). Interestingly, cyclone-induced landslides in Southeast coastal China are clustered in trees-covered slopes, indicating the potential influence of trees on landslide initiation. Here, on the basis of field and experimental evidence, we investigate the impacts of trees on slope stability during a cyclone event in terms of mechanical and hydrological processes.





**Fig. 10** Variation of the factor of safety with depth (1.0–2.0 m below the ground surface)

## 5.1 Mechanical processes of trees

### (1) Root reinforcement

The presence of plant roots can provide additional shear strength for soil to stabilize the slope. Here the direct shear tests indicate that plant roots greatly increase the soil cohesion but weakly contribute to the friction angle, matching the existing results (Genet et al. 2008; Wu 2013; Liang et al. 2015). Due to the limitation of experimental instruments, also to protect the trees, coarse taproots are not involved in the sampling. However, additional cohesion attributable to fine roots reaches 2.2–5.3 kPa, which is close to the effective soil cohesion (Fig. 7e). This additional cohesion plays an important role in holding the soil in place in a cyclone event. Notably, the root reinforcement effect is only identified in the rooting area. Roots reinforce the slope through the friction with soil, thus this beneficial mechanical effect disappears in the rootless area (sampling depths > 0.8 m in laboratory tests). This point is supported by many of the existing experiments and theories, in terms of different types of tree species (Roering et al. 2003; Wu 2013; McGuire et al. 2016).

### (2) Wind load

Scientists recognize that the wind load can act as a driving force to destabilize the slope (Emadi-Tafti and Ataie-Ashtiani 2019). A single tree can transmit wind loads exceeding 10 kN to the slope (Zhuang et al. 2022). Such intense wind loading has been proven to significantly reduce the slope's factor of safety. Though the identified cyclone-induced landslides are located within the radii of seven-scale and ten-scale wind circles, the similarity in the impact area of both wind and rainfall renders their effects on slope stability challenging to differentiate. In this study, the characteristics and spatial distribution of cyclone-induced landslides provide the opportunity to recognize the wind effect.

Cyclone-induced landslides in the study area are clustered in trees/bamboo-covered slopes (Fig. 1). Compared with shrubs and grass, trees have a stronger root system and thus can provide higher additional strength to stabilize the slope. Nevertheless, the wind velocity normally increases with height (according to the wind profile, Mayhoub and Azzam 1997) and the wind load primarily acts on the tree crown to transmit to the slope. During cyclones, trees and bamboo with a large frontal area and high-rising structure can thus transmit greater wind loads to fail the slope.

In terms of geomorphological and geological features, 83.2% of the identified cyclone-induced landslides occur in slopes of a planar and convex upward topography (nose). Only a few landslides initiate in a concave (hollow) topography (Fig. 2d). Existing investigations suggest rainfall-induced landslides are clustered in concave topographies (Reneau et al. 1990; D'Odorico and Fagherazzi 2003; Acharya et al. 2016; Vessia et al. 2020). Due to the low-lying topography, concave slopes normally have a high groundwater table, and tend to maintain a higher moisture content compared to convex slopes. During rainstorms, rainwater infiltrates follow the slope and hydrological gradient. This process results in water concentration in concave areas. Furthermore, field and experimental results of Hales et al. (2009) indicate that additional cohesion attributable to roots is related to topography. Tree roots developed in concave slopes have lower cellulose content and smaller strength compared to roots in convex areas. The weaker roots and water concentration effects make rainfall-induced landslides primarily occur in concave areas. For cyclone-induced landslides, planar and convex upward (nose) slopes are less protected from wind loads (Hales et al. 2009). This implies trees covered on slopes are exposed in the wind field and therefore the wind load is more influential to slope stability than concave scenarios.

The relationship between landslide frequency and slope aspect again illustrates the potential effect of wind loads. Among the 95 identified landslides, 66.4% of the events have a slope aspect of 90°–270°, representing the southern direction (Fig. 2b). In the study area, cyclones form in the Pacific Ocean and propagate from southeast coastal areas to the inland. Therefore, southern slopes are more likely to experience wind loads. The special spatial distribution of identified landslide cases (vegetation cover type, geomorphological and geological features) suggests the wind load is influential to the landslide initiation in the cyclones.

## 5.2 Hydrological processes of trees

### (1) Beneficial hydrological processes

The canopy rainfall interception and transpiration are two primary beneficial hydrological processes of trees. The interception effect decreases the rainwater that penetrates into the slope. However, this process is heavily dependent on the tree species and meteorological conditions. The total interception loss is constrained by the canopy storage capacity (Cisneros Vaca et al. 2018); this threshold is easily exceeded in intense rainfall. For the transpiration effect, its contribution is again related to the meteorological conditions. The daily average transpiration of trees is normally  $< 3$  mm (Lagergren and Lindroth 2002; Ewers et al. 2002), a very small value. In Southeast coastal China, the intensity of cyclone-associated rainfall can reach over 100 mm/d (Qiu et al. 2019; Zhuang et al. 2023). In such a heavy rainstorm, the contribution of beneficial hydrological processes will disappear and weakly affect slope stability.

## (2) Detrimental hydrological processes

Preferential flow attributable to roots, which accelerates rainwater movement within the slope, is known as a detrimental hydrological process of trees (Ghestem et al. 2011). In this study, laboratory tests and dye tracer tests illustrate the great importance of preferential flow arising from tree roots. Laboratory test results suggest the plant roots change the soil porosity and further enlarge the soil permeability. The permeability of root-soil composites reaches 10 times rootless soil (Fig. 7f). Furthermore, dye tracer test results indicate that the contribution of preferential infiltration is several times the shallow uniform infiltration. For the profile excavated in this study, large pores and wormholes are rarely observed. Preferential flow primarily arises from tree roots, more precisely, gaps between root-soil interfaces. There is a positive correlation between roots and dye coverage ( $DC$ ). Dye solution penetrates radially on the roots, thus the dyed area is related to the number and cross-section of roots (Fig. 8). According to existing experiments, preferential flow results from roots is heavily dependent on the root type (Cui et al. 2019; Li et al. 2023). Horizontal roots favor the lateral water penetration in the shallow layer, while the vertical roots facilitate the infiltration of water into the interior of the slope. *Pinus massoniana* in the study area has a taproot system (Yao and Wang 2020; Yang et al. 2021) with a maximum rooting depth of 2 m. Compared with grass and plate-like root systems, coarse taproots provide deep infiltration channels, making rainwater penetrate into the slope within a short duration. Therefore, for the dyed profiles excavated in this study, the penetration depth of preferential flow is much higher than uniform infiltration, and the coarse tree roots appear to be more influential. In addition to root structure, stemflow further enhances this process by concentrating rainwater at the base of tree trunks, thereby increasing water input near root zones. This localized influx intensifies infiltration through root-associated preferential pathways, particularly along root-soil interfaces. The combination of stemflow and deep taproots forms an efficient vertical conduit for water to penetrate into the slope interior within a short time. These findings highlight that preferential flow driven by both stemflow and root systems should be considered when conducting stability analysis of trees-covered slopes in cyclones.

## 5.3 Contributions of trees to initiation of cyclone-induced landslides

Cyclone-induced landslides in the study area primarily occur at the bedrock-soil interface and the possible weak layer in the overburden soil that is close to the maximum rooting

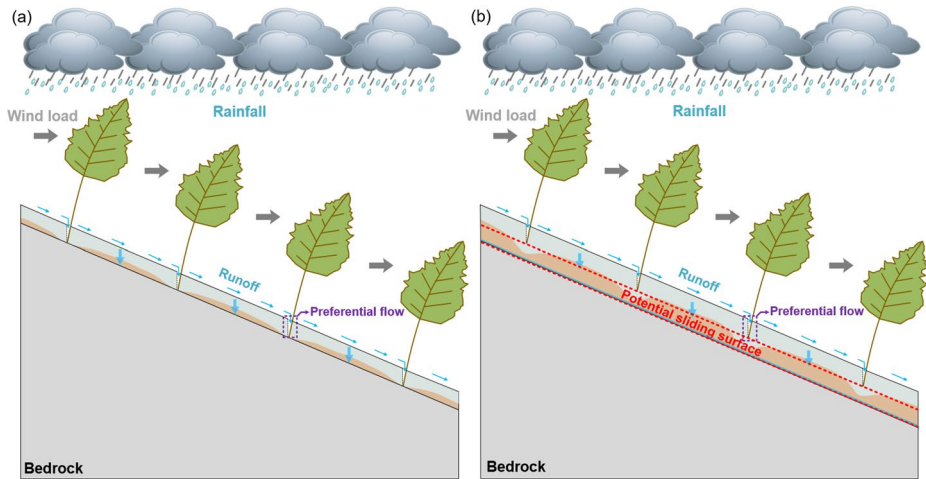
depth (Zhuang et al. 2022). They happen at approximately the same time as cyclone strikes, showing a weak time-lag effect. The mechanical and hydrological processes attributable to trees provide explanations for the landslide initiation.

Our test results indicate that reinforcement and preferential flow arising from tree roots change the soil properties, and they are both heavily related to the maximum rooting depth. Cohesion attributable to roots is only available within the root-reinforced area, leading to a strength difference at the maximum rooting depth. Moreover, a remarkable drop of permeability is identified at the same place. Preferential flow paths provided by tree roots accelerate water infiltration and make the shallow reinforced soil of high moisture in a cyclone. Rainwater will accumulate at the end of root channels, forming regional high pore water pressure (Ghestem et al. 2011; Guo et al. 2020). Both mechanical and hydrological processes make the maximum rooting depth a weak plane. Regarding trees transmit wind loads to slopes through root systems, slope failure will occur when the total driving force (wind load, soil gravity and tree surcharge) exceeds the resistance. Therefore, the maximum rooting depth becomes a potential sliding surface in a cyclone event.

For slopes fail at bedrock-soil interfaces, the property difference between soil and bedrock plays an important role. The bedrock of high integrity can be treated as an impermeable layer of high strength. When rainwater reaches the bedrock-soil interface, lateral infiltration will happen. A transient saturation zone then forms in the area, decreasing the matric suction and shear strength of the overburden rootless soil (Lanni et al. 2012). Therefore, the bedrock-soil interfaces become another potential sliding surface when the slope subject to wind load and intense rainfall. Compared with cases that occur near the maximum rooting depth, the influence of trees on slope stability is relatively weak in this scenario. However, cyclones landed in the study area mostly have a short duration (Zhuang et al. 2022). We suggest the preferential flow arising from tree roots plays an important role in destabilizing the slope during a short-duration cyclone.

For cyclone-induced landslides in Southeast coastal China, the depth of soil layer is mostly larger than the rooting depth. Tree roots hinder the development of sliding surfaces within the reinforced area, but make the maximum rooting depth and bedrock-soil interfaces two potential sliding surfaces. However, when the soil depth is smaller than the rooting depth, the root reinforcement process appears to be more influential. The additional shear strength attributable to roots can resist the driving force of wind loads, holding the soil in place (Zhuang et al. 2022). Cyclone-induced landslides in New Zealand support our conclusion. Reid and Page (2002) investigated cyclone Bola-induced landslides in New Zealand. Soil layer in the area is shallow with a depth of 0–2 m, and 55% of landslides are 0.5–0.9 m in depth. Trees (*Pinus Radiata*) covered on slopes have a maximum rooting depth of ~1.8 m (Waston et al. 1999), which can penetrate the soil layer. The root reinforcement remains the slope stable during cyclones even when the soil is fully saturated. The landslide frequency in trees-covered slopes therefore is 16 times lower in pastures (Marden and Rowan 1994), showing different features of cyclone-induced landslides in Southeast China. Based on this, we preliminarily suggest defining an index  $\alpha = h_r/h_s$  ( $h_r$  and  $h_s$  represent the maximum rooting depth and soil depth, respectively) to clarify the dominant contribution of trees on slope stability during cyclones. When the root depth  $h_r$  is greater than the soil depth  $h_s$ , tree roots reinforce the entire soil layer and resist the detrimental mechanical process of wind loads, stabilizing the slope during cyclones (Fig. 11a). When the root depth  $h_r$  is smaller than the soil depth  $h_s$ , tree-stabilizing effects are limited. The reinforced soil is stabilized. However,





**Fig. 11** Impacts of trees on slope stability during cyclones, indicating the location of potential sliding surfaces. (a) Root depth  $h_r >$  soil depth  $h_s$ ; (b) Root depth  $h_r <$  soil depth  $h_s$

the detrimental processes of wind load and preferential flow favor the formation of two potential sliding surfaces: maximum rooting depth and bedrock-soil interface (Fig. 11b).

The impact of trees on slope stability is heavily dependent on geological settings of the slope and meteorological conditions. For cyclone-induced landslides, in some cases, the detrimental impacts of preferential flow and wind loads can outweigh the impacts of root reinforcement and beneficial hydrological processes. Here we underscore the main result that cyclone-induced landslides in Southeast coastal China are jointly triggered by the wind load, intense rainfall and trees. The dominant mechanisms of trees on slope stability have to be carefully clarified on the basis of the geological and meteorological conditions, rather than overestimate the stabilization effects.

## 6 Conclusions

Cyclone-induced landslides in Southeast coastal China mainly develop in trees-covered slopes, showing a potential relationship of failure mechanism with trees. To elucidate the influence of trees on slope stability, we investigated 95 cyclone-induced landslides in Southeast coastal China and summarized their general features. Further field tests, laboratory tests and dye tracer tests were conducted to investigate the tree impacts in terms of mechanical and hydrological processes. Our works clarify the dominant mechanism of trees on slope stability during cyclones and suggest the necessity of including tree impacts when conducting landslide risk assessments. The main conclusions are drawn as follows:

- (1) Cyclone-induced landslides in the study area are clustered in trees-covered, south-facing slopes of a planar and convex upward (nose) topography. The special spatial distribution and characteristics of identified landslides confirm the contribution of wind loads on slope failure in cyclone events.

- (nose) topography. The special spatial distribution and characteristics of identified landslides confirm the contribution of wind loads on slope failure in cyclone events.
- (2) *Pinus massoniana*, which is widely distributed in the study area, has a strong taproot system. The maximum rooting depth reaches 2 m and lateral roots are relatively sparse. At depths > 0.4 m, roots developed in the slope primarily attributable to trees and the contribution of fern (*Pteris multifida*) is weak. Compared with fine roots, the *RAD* proportion of coarse tree roots accounts for over 80% in the entire profile, playing a vital role in affecting slope stability.
  - (3) The presence of tree roots changes the mechanical and hydrological properties of soil. Tree roots provide additional cohesion to reinforce the slope, but also magnify soil porosity and create preferential flow channels. The permeability of root-soil composites reaches 10 times rootless soil. Furthermore, dye tracer tests indicate that preferential flow paths in trees-covered slopes primarily attribute to the gaps between root-soil interfaces. The proportion of preferential infiltration in the two dyed profiles is 2–3 times shallow uniform infiltration, thus the primary source of soil moisture in cyclones.
  - (4) The impacts of trees on slope stability are heavily dependent on slope settings and meteorological conditions. The beneficial hydrological processes of rainfall interception and transpiration almost disappear in rainstorms, while the root reinforcement and preferential flow are related to the depths of tree roots and soil layer. When the soil depth is larger than the maximum rooting depth, the detrimental impacts of wind loads and preferential infiltration can exceed the stabilizing effect of root reinforcement. The maximum rooting depth and bedrock-soil interfaces become potential sliding surfaces when trees-covered slopes subject to strong wind load and rainfall in cyclones.

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**Author contributions** YZ analyzed the results and wrote the paper; AX designed the research; All the authors did the field investigation, did the experiments and edited the paper.

## Declarations

**Conflict of interest** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## References

- Acharya KP, Bhandary NP, Dahal RK, Yatabe R (2016) Seepage and slope stability modelling of rainfall-induced slope failures in topographic hollows. *Geomatics Nat Hazards Risk* 7(2):721–746
- Ali W, Hussain S, Chen J, Hu F, Liu J, He Y, Yang M (2023) Cover crop root-derived organic carbon influences aggregate stability through soil internal forces in a clayey red soil. *Geoderma* 429:116271
- Allaire SE, Roullet S, Cessna AJ (2009) Quantifying Preferential flow in soils: a review of different techniques. *J Hydrol* 378:179–204
- Bargués Tobella A, Reese H, Almaw A, Bayala J, Malmer A, Laudon H, Ilstedt U (2014) The effect of trees on Preferential flow and soil infiltrability in an agroforestry parkland in semiarid Burkina Faso. *Water Resource Res* 50:3342–3354
- Bodner G, Leitner D, Kaul HP (2014) Coarse and fine root plants affect pore size distributions differently. *Plant Soil* 380:133–151

- Chang KT, Chiang SH, Chen YC, Mondini AC (2014) Modeling the Spatial occurrence of shallow landslides triggered by typhoons. *Geomorphology* 208:137–148
- Cislaghi A, Rigon E, Lenzi MA, Bischetti GB (2018) A probabilistic multidimensional approach to quantify large wood recruitment from hillslopes in mountainous-forested catchments. *Geomorphology* 306:108–127
- Cislaghi A, Cohen D, Gasser E, Bischetti GB, Schwarz M (2019) Field measurements of passive Earth forces in steep, shallow, landslide-prone areas. *J Geophys Research: Earth Surf* 124(3):838–866
- Cisneros Vaca C, van der Tol C, Ghimire CP (2018) The influence of long-term changes in canopy structure on rainfall interception loss: a case study in Speulderbos, the Netherlands. *Hydrol Earth Syst Sci* 22(7):3701–3719
- Cogan J, Gratchev I, Wang GH (2018) Rainfall-induced shallow landslides caused by ex-Tropical cyclone Debbie, 31st March 2017. *Landslides* 15:1215–1221
- Cohen D, Schwarz M (2017) Tree-root control of shallow landslides. *Earth Surf Dyn* 5:451–477
- Cui Z, Wu GL, Huang Z, Liu Y (2019) Fine roots determine soil infiltration potential than soil water content in semi-arid grassland soils. *J Hydrol* 578:124023
- D'Odorico P, Fagherazzi S (2003) A probabilistic model of rainfall-triggered shallow landslides in hollows: A long-term analysis. *Water Resour Res* 39:1262
- Deljouei A, Cislaghi A, Abdi E, Borz SA, Majnounian B, Hales TC (2023) Implications of Hornbeam and Beech root systems on slope stability: from field and laboratory measurements to modelling methods. *Plant Soil* 483:547–572
- Elsner JB, Kossin JP, Jagger TH (2008) The increasing intensity of the strongest tropical cyclones. *Nature* 455(7209):92–95
- Emadi-Tafti M, Ataie-Ashtiani B (2019) A modeling platform for landslide stability: A hydrological approach. *Water* 11(10):2146
- Ewers BE, Mackay DS, Gower ST, Ahl DE, Burrows SN, Samanta SS (2002) Tree species effects on stand transpiration in Northern Wisconsin. *Water Resour Res* 38(7):1–11
- Genet M, Kokutse N, Stokes A, Fourcaud T, Cai XH, Ji JN, Mickovski S (2008) Root reinforcement in plantations of cryptomeria Japonica D. Don: effect of tree age and stand structure on slope stability. *For Ecol Manag* 256:1517–1526
- Ghestem M, Sidle RC, Stokes A (2011) The influence of plant root systems on subsurface flow: implications for slope stability. *Bioscience* 61(11):869–879
- Gonzalez-Ollauri A, Mickovski SB (2017) Hydrological effect of vegetation against rainfall-induced landslides. *J Hydrol* 549:374–387
- Guo L, Liu Y, Wu GL, Huang Z, Cui Z, Cheng Z, Zhang RQ, Tian FP, He HH (2019) Preferential water flow: influence of alfalfa (*Medicago sativa* L.) decayed root channels on soil water infiltration. *J Hydrol* 578:124019
- Guo WZ, Chen ZX, Wang WL, Gao WW, Guo MM, Kang HL, Li PF, Wang WX, Zhao M (2020) Telling a different story: the promote role of vegetation in the initiation of shallow landslides during rainfall on the Chinese loess plateau. *Geomorphology* 350:106879
- Hales TC, Ford CR, Hwang T, Vose JM, Band LE (2009) Topographic and ecologic controls on root reinforcement. *J Phys Res* 114(F3):F03013
- Huang MF, Wang Q, Liu MF, Lin N, Wang YF, Jing RZ, Sun JP, Murakami H, Lou WJ (2022) Increasing typhoon impact and economic losses due to anthropogenic warming in Southeast China. *Sci Rep* 12:14048
- Jotisankasa A, Sirirattanachai T (2017) Effects of grass roots on soil-water retention curve and permeability function. *Can Geotech J* 54(11):1612–1622
- Kramers G, Richards KG, Holden NM (2009) Assessing the potential for the occurrence and character of Preferential flow in three Irish grassland soils using image analysis. *Geoderma* 153(3–4):362–371
- Lagergren F, Lindroth A (2002) Transpiration response to soil moisture in pine and Spruce trees in Sweden. *Agric for Meteorol* 112(2):67–85
- Lann T, Bao H, Lan H, Zheng H, Yan C, Peng J (2024) Hydro-mechanical effects of vegetation on slope stability: A review. *Sci Total Environ* 926:171691
- Lanni C, McDonnell J, Hopp L, Rigon R (2012) Simulated effect of soil depth and bedrock topography on near-surface hydrologic response and slope stability. *Earth Surf Proc Land* 38(2):146–159
- Li JY, Cui P, Yin YZ (2023) Field observation and micro-mechanism of roots-induced Preferential flow by infiltration experiment and phase-field method. *J Hydrol* 623:109756
- Liang T, Knappett JA, Duckett N (2015) Modelling the seismic performance of rooted slopes from individual root-soil interaction to global slope behaviour. *Géotechnique* 65(12):995–1009
- Luo ZT, Niu JZ, Zhang L, Chen XW, Zhang W, Xie BY, Du J, Zhu ZJ, Wu SS, Li X (2019) Roots-Enhanced Preferential flows in deciduous and coniferous forest soils revealed by Dual-Tracer experiments. *J Environ Qual* 48(1):136–146

- Marden M, Rowan D (1994) Protective value of vegetation on tertiary terrain before and during cyclone Bola, East Coast, North Island, new Zealand. *N Z J Forest Sci* 23:255–263
- Masi EB, Segoni S, Tofani V (2021) Root reinforcement in slope stability models: a review. *Geosciences* 11(5):212
- Mayhoub AB, Azzam A (1997) A survey on the assessment of wind energy potential in Egypt. *Renewable Energy* 11(2):235–247
- McGuire LA, Rengers FK, Kean JW, Coe JA, Mirus BB, Baum RL, Godt JW (2016) Elucidating the role of vegetation in the initiation of rainfall-induced shallow landslides: insights from an extreme rainfall event in the Colorado front range. *Geophys Res Lett* 43:9084–9092
- Mills A (2012) An overview of the methylene blue ISO test for assessing the activities of photocatalytic films. *Appl Catal B* 128:144–149
- Nandy SK, Venkatesh KV (2010) Application of methylene blue dye reduction test (MBRT) to determine growth and death rates of microorganisms. *Afr J Microbiol Res* 4(1):061–070
- Ng CW, W, Kamchoom V, Leung AK (2016) Centrifuge modelling of the effects of root geometry on transpiration-induced Suction and stability of vegetated slopes. *Landslides* 13:925–938
- Nimmo JR (2021) The processes of Preferential flow in the unsaturated zone. *Soil Sci Soc Am J* 85(1):1–27
- Ohlmacher GC (2007) Plan curvature and landslide probability in regions dominated by Earth flows and Earth slides. *Eng Geol* 91(2–4):117–134
- Patricola CM, Wehner MF (2018) Anthropogenic influences on major tropical cyclone events. *Nature* 563:339–346
- Qiu WY, Ren FM, Wu LG, Chen LS, Ding CC (2019) Characteristics of tropical cyclone extreme precipitation and its preliminary causes in Southeast China. *Meteorol Atmos Phys* 131(3):613–626
- Rasband WS (2007) ImageJ. U.S. National institute of Health
- Reid LM, Page MJ (2002) Magnitude and frequency of landsliding in a large new Zealand catchment. *Geomorphology* 49(1–2):71–88
- Reneau SL, Dietrich WE, Donahue DJ, Jull AJT, Rubin M (1990) Late quaternary history of colluvial deposition and erosion in hollows, central California Coast ranges. *Geol Soc Am Bull* 102:969–982
- Roering JJ, Schmidt KM, Stock JD, Dietrich WE, Montgomery DR (2003) Shallow landsliding, root reinforcement, and the Spatial distribution of trees in the Oregon Coast range. *Can Geotech J* 40(2):237–253
- Saito H, Korup O, Uchida T, Hayashi S, Oguchi T (2014) Rainfall conditions, typhoon frequency, and contemporary landslide erosion in Japan. *Geology* 42(11):999–1002
- Sandoval GF, Galobardes I, Teixeira RS, Toralles BM (2017) Comparison between the falling head and the constant head permeability tests to assess the permeability coefficient of sustainable pervious concretes. *Case Stud Constr Mater* 7:317–328
- Schwarz M, Giadrossich F, Cohen D (2013) Modeling root reinforcement using a root-failure Weibull survival function. *Hydrol Earth Syst Sci* 17:4367–4377
- Sidle RC, Ziegler AD (2017) The canopy interception–landslide initiation conundrum: insight from a tropical secondary forest in Northern Thailand. *Hydrol Earth Syst Sci* 21(1):651–667
- Simon A, Collison AJ (2002) Quantifying the mechanical and hydrologic effects of riparian vegetation on streambank stability. *Earth Surf Proc Land* 27(5):527–546
- Song X, Tan Y, Lu Y (2024) Microscopic analyses of the reinforcement mechanism of plant roots in different morphologies on the stability of soil slopes under heavy rainfall. *CATENA* 241:108018
- USDA (2004) Soil survey laboratory methods manual. Soil survey investigations report no. 42, ver. 4.0. US Department of Agriculture, Natural Resource Conservation Service, Washington, DC
- Vessia G, Curzio DD, Chiaudani A, Rusi S (2020) Regional rainfall threshold maps drawn through multivariate Geostatistical techniques for shallow landslide hazard zonation. *Sci Total Environ* 705:135815
- Wang ZC (2014) Study on the effect of vegetation on slope stability under the wind load. Chengdu University of Technology
- Waston A, Phillips C, Marden M (1999) Root strength, growth, and rates of decay: root reinforcement changes of two tree species and their contribution to slope stability. *Plant Soil* 217:39–47
- Wu TH (2013) Root reinforcement of soil: review of analytical models, test results, and applications to design. *Can Geotech J* 50(3):259–274
- Xiao YF, Duan ZD, Xiao YQ, Ou JP, Chang L, Li QS (2011) Typhoon wind hazard analysis for Southeast China coastal regions. *Struct Saf* 33(4–5):286–295
- Yang ZY, Zhou BZ, Ge XG, Cao YH, Brunner I, Shi JX, Li MH (2021) Species-specific responses of root morphology of three co-existing tree species to nutrient patches reflect their root foraging strategies. *Front Plant Sci* 11:618222
- Yao RL, Wang Y (2020) An advanced protocol for the establishment of plantlets originating from somatic embryos in *Pinus massoniana*. *3 Biotech* 10:394
- Zhang TL (2016) The research of deformation characteristics and formation mechanism of landslides induced by typhoon rainfall in the Eastern part of Zhejiang Province. China University of Geosciences

- Zhang JM, Xu ZM (2016) Dye tracer infiltration technique to investigate macropore flow paths in Maka Mountain, Yunnan Province, China. *J Cent South Univ* 23:2101–2109
- Zhu H, Zhang LM (2015) Evaluating Suction profile in a vegetated slope considering uncertainty in transpiration. *Comput Geotech* 63:112–120
- Zhuang Y, Xing AG, Jiang YH, Sun Q, Yan JK, Zhang YB (2022) Typhoon, rainfall and trees jointly cause landslides in coastal regions. *Eng Geol* 298:106561
- Zhuang Y, Xing AG, Petley D, Jiang YH, Sun Q, Bilal M, Yan JK (2023) Elucidating the impacts of trees on landslide initiation throughout a typhoon: Preferential infiltration, wind load and root reinforcement. *Earth Surf Proc Land* 48(15):3128–3141

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